

MySQL & MySQL

HeatWave Report

June 2026

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Me, Myself & I

➤ **Data Geek**

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MySQL & MySQL HeatWave 9.7.0 Report

- Welcome to the MySQL and MySQL HeatWave Report – June 2026 edition
- This report covers the most relevant and new features of **MySQL 9.7.0 LTS** & **MySQL HeatWave 9.7.0**, both released in **April 2026**
- MySQL 9.7.0 LTS establishes the new 9.7.x Long-Term Support release line, giving users a stable branch to standardize on while building on the innovation delivered through the 9.x innovation cycle
- This release significantly expands MySQL with new capabilities...

MySQL 9.7.0 LTS – Why it Matters

TL;DR

- For DBAs
 - Better replication visibility
 - Stronger security controls
 - Easier HA management
 - Improved observability
- For Developers
 - JSON Duality Views
 - Better query optimization
 - OpenTelemetry integration
- For CTOs
 - New LTS platform
 - Expanded Community Edition capabilities
 - Lower operational complexity
 - Stronger governance and compliance

MySQL 9.7.0 LTS - A New Milestone

- MySQL 9.7.0 introduces the next Long-Term Support (LTS) release series after MySQL 8.4
- It consolidates innovations delivered during the 9.x cycle and **expands several capabilities previously restricted to Enterprise Edition into Community Edition**
- **Key themes:**
 - Better observability
 - Improved High Availability
 - Modern application development
 - Query optimization
 - Stronger security
- MySQL 9.7 becomes the recommended long-term target for organizations wanting the latest LTS platform

MySQL 9.7.0 LTS - A New Milestone

- **Resources**

- MySQL 9.7 Release Notes

- <https://dev.mysql.com/doc/relnotes/mysql/9.7/en/>

- MySQL 9.7 LTS Announcement

- <https://blogs.oracle.com/mysql/mysql-9-7-0-lts-is-now-available-expanded-community-capabilities-and-dynamic-data-masking-for-enterprise>

Atomic DDL and InnoDB Improvements

Atomic DDL and InnoDB Enhancements

- **Atomic DDL:** Dropping a database is now possible even if its directory is missing from the file system
- **GIPK Improvements:** *UNIQUE NOT NULL* keys are now treated as Primary Key-equivalent for *CREATE* and *ALTER* statements when *sql_generate_invisible_primary_key=ON*
- **InnoDB:** MySQL Clone plugin now supports cloning between consecutive LTS versions higher than 9.7.0

Atomic DDL and InnoDB Enhancements

- **Resources**

- Atomic Data Definition Statement Support
 - <https://dev.mysql.com/doc/refman/9.7/en/atomic-ddl.html>
- Generated Invisible Primary Keys (GIPK)
 - <https://dev.mysql.com/doc/refman/9.7/en/create-table-gipks.html>
- The clone plugin
 - <https://dev.mysql.com/doc/refman/9.7/en/clone-plugin.html>
- The InnoDB Storage Engine
 - <https://dev.mysql.com/doc/refman/9.7/en/innodb-storage-engine.html>

Hypergraph Optimizer

Hypergraph Optimizer (Community Edition)

- The Hypergraph Optimizer is now available in MySQL Community Edition as an alternative to the classic join optimizer
- It improves optimization of complex multi-table joins by evaluating a larger search space than traditional join ordering algorithms
- **Benefits:**
 - Better execution plans
 - Faster analytical queries
 - Improved performance for complex schemas
 - Reduced need for optimizer hints
- This is one of the most significant query optimization enhancements ever delivered to Community Edition

Hypergraph Optimizer (Community Edition)

Example

```
-- Enable hypergraph optimizer for the current session
SET optimizer_switch='hypergraph_optimizer=on';

-- Or enable for a single statement using a hint
SELECT /*+ SET_VAR(optimizer_switch='hypergraph_optimizer=on') */
*
FROM orders o
INNER JOIN customers c ON o.customer_id = c.id
INNER JOIN products p ON o.product_id = p.id
INNER JOIN categories cat ON p.category_id = cat.id
WHERE o.order_date > '2026-04-21';
```

Hypergraph Optimizer (Community Edition)

- **Resources**

- The hypergraph optimizer is now available in MySQL 9.7 Community Edition
 - <https://blogs.oracle.com/mysql/the-hypergraph-optimizer-is-now-available-in-mysql-9-7-community-edition>
- MySQL 9.7 Reference Manual – Switchable Optimizations
 - <https://dev.mysql.com/doc/refman/9.7/en/switchable-optimizations.html>
- MySQL 9.7.0 Release Notes
 - <https://dev.mysql.com/doc/relnotes/mysql/9.7/en/news-9-7-0.html#mysqld-9-7-0-optimizer>

Performance and Resource Awareness

PGO & cpuset cgroup awareness

- **PGO:** Profile-Guided Optimization is extended for 9.7.0 with RPM build support on SLE/openSUSE and Fedora
 - PGO is a compiler optimization technique that uses runtime profiling to produce better-optimized binaries, targeting improved performance for real-world workloads
- **Resource-Aware Deployments:** MySQL Server now supports handling resource limitations set using the cpuset cgroup, improving the calculation of available logical CPUs in constrained environments (like containers)
 - Particularly useful for Kubernetes and managed cloud environments

PGO & cpuset cgroup awareness

- **Resources**

- MySQL 9.7.0 Release Notes (PGO)
 - <https://dev.mysql.com/doc/relnotes/mysql/9.7/en/news-9-7-0.html#mysqld-9-7-0-packaging>
- MySQL 9.7.0 Release Notes (cgroup awareness)
 - <https://dev.mysql.com/doc/relnotes/mysql/9.7/en/news-9-7-0.html#mysqld-9-7-0-innodb>
- WL#16851: Support cpuset-cpus cgroup
 - <https://dev.mysql.com/worklog/task/?id=16851>

JSON Duality Views

JSON Duality Views

- JSON Duality Views are available in MySQL Community Edition
 - With support of DML operations (insert, update, delete)
 - Auto-increment support is also added for easier data insertion with generated keys
- They provide a unified representation of relational & document data
- **Benefits:**
 - Single source of truth
 - JSON API friendly
 - Relational consistency
 - Easier application development
- Developers can work with JSON documents while preserving normalized relational storage

JSON Duality Views

Example

```
-- Define a JSON Duality View
CREATE JSON RELATIONAL DUALITY VIEW customer_orders_dv AS
  SELECT JSON_DUALITY_OBJECT(
    '_id': customer_id,
    'customer_name': name,
    'orders': (
      SELECT JSON_ARRAYAGG(
        JSON_DUALITY_OBJECT(
          'order_id': order_id,
          'product': product,
          'amount': amount
        )
      )
    )
  FROM orders WHERE orders.customer_id = customers.customer_id
)
FROM customers;

SELECT JSON_PRETTY(data) FROM customer_orders_dv WHERE data->'$_id' = 1;
```

JSON Duality Views

- **Resources**

- MySQL 9.7 Reference Manual – JSON Duality Views
 - <https://dev.mysql.com/doc/refman/9.7/en/json-duality-views.html>
- MySQL 9.7 Reference Manual – Creating JSON Duality Views
 - <https://dev.mysql.com/doc/refman/9.7/en/json-duality-views-syntax.html>
- Two Models, One World: Introducing JSON Relational Duality Views in MySQL HeatWave
 - <https://blogs.oracle.com/mysql/two-models-one-world-introducing-json-relational-duality-views-in-mysql-heatwave>
- Hands-on with JSON Duality Views in MySQL: From Creation to Day-to-Day Use
 - <https://blogs.oracle.com/mysql/defining-json-duality-views-in-mysql>
- Oracle blog announcement
 - <https://blogs.oracle.com/mysql/mysql-9-7-0-its-is-now-available-expanded-community-capabilities-and-dynamic-data-masking-for-enterprise>

OpenTelemetry

OpenTelemetry Integration

- Native OpenTelemetry (Otel) support arrives in Community Edition
 - Enabling export of logs, metrics, and traces to OpenTelemetry backends for centralized observability
- This component-based integration allows MySQL to fit seamlessly into modern observability pipelines, making it easier for DBAs, SREs, and platform teams to monitor MySQL alongside the rest of their stack
- **Benefits:**
 - Standardized metrics
 - Distributed tracing integration
 - Easier cloud-native monitoring
 - Better troubleshooting
- Particularly useful for Kubernetes and microservices environments

OpenTelemetry Integration

Example

```
-- Install the OpenTelemetry server component
INSTALL COMPONENT 'file://component_telemetry';

-- View telemetry-specific system variables
SHOW VARIABLES LIKE '%telemetry%';
```

OpenTelemetry Integration

- **Resources**

- MySQL 9.7 Reference Manual – Telemetry
 - <https://dev.mysql.com/doc/refman/9.7/en/telemetry.html>
- MySQL 9.7 Reference Manual – OpenTelemetry Logging
 - <https://dev.mysql.com/doc/refman/9.7/en/telemetry-logging.html>
- OpenTelemetry Documentation
 - <https://opentelemetry.io/docs/what-is-opentelemetry/>

Replication Enhancements

Replication Observability & HA Enhancements

- MySQL 9.7 brings significant replication observability and high availability improvements across four key areas:
 - **Flow-control monitoring** – Better visibility into when a cluster is being throttled and how much impact that throttling is having
 - **Extended applier statistics** – Better visibility into lag, throughput, queued work, and worker behavior in multi-threaded replication
 - **Automatic Eviction & Rejoin** – More resilient cluster behavior through detection of unhealthy members, eviction of unstable nodes, and automatic recovery when conditions improve
 - **Up-to-date Aware Primary Election** – Better failover quality by preferring the most current eligible member during primary election
- **Benefits:**
 - Faster troubleshooting
 - Better HA operations
 - Improved scaling decisions

Replication Observability & HA Enhancements

- **Resources**

- MySQL Replication Monitoring – Enhanced Features
 - <https://blogs.oracle.com/mysql/mysql-replication-monitoring-enhanced-features-for-the-enterprise-edition>
- MySQL 9.7 Reference Manual – Replication
 - <https://dev.mysql.com/doc/refman/9.7/en/replication.html>
- MySQL 9.7 Reference Manual – Group Replication
 - <https://dev.mysql.com/doc/refman/9.7/en/group-replication.html>
- Replication Components
 - <https://dev.mysql.com/doc/refman/9.7/en/replication-components.html>
- MySQL 9.7 Reference Manual – MySQL InnoDB Cluster
 - <https://dev.mysql.com/doc/mysql-shell/9.7/en/mysql-innodb-cluster.html>

Stronger Password Security

Authentication – PBKDF2 with SHA512

- Authentication now supports PBKDF2 storage format with `cached_sha2_password`, providing enhanced security and flexibility
- This update allows for the use of `PBKDF2` with `SHA512`, enabling a smoother migration path from existing formats and ensuring even stronger security guarantees
- **Benefits:**
 - Stronger password protection
 - Better resistance against offline attacks
 - Modernized authentication security

Authentication – PBKDF2 with SHA512

- **Resources**

- MySQL 9.7 Release Notes – Authentication
 - <https://dev.mysql.com/doc/relnotes/mysql/9.7/en/news-9-7-0.html#mysqld-9-7-0-authentication>
- Caching SHA-2 Pluggable Authentication
 - <https://dev.mysql.com/doc/refman/9.7/en/caching-sha2-pluggable-authentication.html>

Dynamic Data Masking

Dynamic Data Masking (Enterprise Edition)

- Available in MySQL 9.7 Enterprise Edition & MySQL HeatWave
- You attach a masking policy directly to a base-table column so that MySQL returns either the original value or a masked value at query time, based on the executing user or active role, without application changes and without maintaining separate masked copies of data
- This helps organizations implement least-privilege access and reduce sensitive-data exposure for compliance with GDPR, HIPAA, PCI DSS, SOX, and similar standards
- **Benefits:**
 - No application changes
 - Centralized security
 - Reduced exposure of PII
 - Easier regulatory compliance

Dynamic Data Masking (Enterprise Edition)

Example 1/2

```
-- Install the Enterprise object_policy component
SOURCE install_component_object_policy.sql;

-- Grant this to the user who will create, show, or drop masking policies
GRANT MANAGE_DATA_MASKING_POLICY ON *.* TO '<mask_admin>'@'%';

-- Create a table with sensitive data
CREATE TABLE user_profiles (    id INT PRIMARY KEY, first_name VARCHAR(32),
last_name VARCHAR(32), ssn VARCHAR(16), zip_code VARCHAR(10));

-- Setup identities: users and a role
CREATE USER IF NOT EXISTS 'seeall'@'%' IDENTIFIED BY 'Welcome_123!';
CREATE USER IF NOT EXISTS 'noseeall'@'%' IDENTIFIED BY 'Welcome_123!';

GRANT SELECT ON protected.user_profiles TO 'seeall'@'%';
GRANT SELECT ON protected.user_profiles TO 'noseeall'@'%';
```

Dynamic Data Masking (Enterprise Edition)

Example 2/2

```
-- Setup identities: users and a role
CREATE ROLE IF NOT EXISTS 'pii_read'@'%';
GRANT SELECT ON protected.user_profiles TO 'pii_read'@'%';

GRANT 'pii_read'@'%' TO 'seeall'@'%';

-- Create a masking policy to mask SSNs (user-based)
CREATE MASKING POLICY mask_ssn_policy(ssn_col)
  CASE WHEN CURRENT_USER_IN('seeall')
    THEN ssn_col
    ELSE mask_ssn(ssn_col)
  END;

-- Apply the policy to the SSN column
ALTER TABLE protected.user_profiles
  ALTER COLUMN ssn
  SET MASKING POLICY mask_ssn_policy;
```

*As noseeall, SSNs are masked
(e.g., XXX-XX-0001)*

*As seeall (and/or with the appropriate
active role), SSNs are visible*

Dynamic Data Masking (Enterprise Edition)

- **Resources**

- Dynamic Data Masking in MySQL – Protect Sensitive Data Without App Changes
 - <https://blogs.oracle.com/mysql/dynamic-data-masking-in-mysql-protect-sensitive-data-without-app-changes>
- MySQL 9.7 Reference Manual – MySQL Enterprise Data Masking
 - <https://dev.mysql.com/doc/refman/9.7/en/data-masking.html>

Audit Log Enhancements

Audit Log Enhancements (Enterprise Edition)

- MySQL 9.7 introduces several audit log improvements:
 - **Time-based rotation** – The audit log can now be rotated based on elapsed time, controlled by *audit_log.rotate_on_time*.
 - **Automatic filter recovery** – If an invalid filter configuration is detected at startup, it is replaced in-memory with a default filter that logs all events, preventing loss of audit activity. This behavior is controlled by *audit_log_filter_recovery_mode*
3 recovery modes are available:
 - *LOG_ALL_IF_INVALID_FILTER_DETECTED*
 - *LOG_NOTHING_IF_INVALID_FILTER_DETECTED*
 - *ABORT_IF_INVALID_FILTER_DETECTED*
 - **Relaxed gzip header checks** – The server now allows ordinary gzip files to be processed.

Audit Log Enhancements (Enterprise Edition)

- **Resources**

- MySQL 9.7 Reference Manual – Audit Log Component
 - <https://dev.mysql.com/doc/refman/9.7/en/audit-log-component.html>
- MySQL 9.7 Release Notes – Audit Log
 - <https://dev.mysql.com/doc/relnotes/mysql/9.7/en/news-9-7-0.html#mysqld-9-7-0-audit-log>

MySQL HeatWave Replica Clone

MySQL HeatWave Replica Clone

- MySQL HeatWave on OCI introduces a simplified workflow for creating replica DB Systems
- It allows cloning within the same region or across regions, while optionally configuring replication automatically
- This simplifies Disaster Recovery setups by orchestrating backups, cross-region copies, restores, and replication channel creation behind the scenes

Benefits:

- Faster disaster recovery deployments
- Easier read scaling
- Simplified replication setup
- Reduced operational effort

MySQL HeatWave Replica Clone

- Typical Use Cases
 - Reporting replicas
 - Cross-region DR
 - Development environments
 - Read scaling (in another region)
 - If in the same region, use the Read Replica feature instead

MySQL HeatWave Replica Clone

- **Resources**

- Create Replica DB system Made Easy for MySQL HeatWave Service on OCI
 - <https://blogs.oracle.com/mysql/create-replica-db-system-made-easy-for-mysql-heatwave-service-on-oci>
- Creating a DB System Clone
 - <https://docs.oracle.com/en-us/iaas/mysql-database/doc/cloning-db-system.html>

Bulk Loading with Foreign Keys for MySQL HeatWave

MySQL HeatWave – Bulk Loading with FK

- MySQL HeatWave now supports bulk loading of tables with foreign keys, ensuring that all foreign key constraints are satisfied for each record
- This update enables you to successfully load data into tables with foreign key constraints while providing error reporting for violated constraints

MySQL HeatWave – Bulk Loading with FK

- **Resources**

- MySQL HeatWave Release Notes – Bulk Loading

- <https://dev.mysql.com/doc/relnotes/heatwave/en/news-9-7-0.html#heatwave-9-7-0-hw-mysql>

- MySQL HeatWave Documentation – Bulk Ingest Data

- <https://dev.mysql.com/doc/heatwave/en/mys-hw-bulk-load-mysql-server.html>

MySQL HeatWave GenAI NL_SQL with Gemini and OpenAI

HeatWave GenAI – NL_SQL with Gemini & OpenAI

- The NL_SQL routine in MySQL HeatWave GenAI now supports Gemini and OpenAI models
- This enables natural language to SQL translation using a broader range of large language models, giving users more flexibility in choosing the AI model that best fits their needs and preferences

HeatWave GenAI – NL_SQL with Gemini & OpenAI

- **Resources**

- MySQL HeatWave Release Notes – GenAI
 - <https://dev.mysql.com/doc/relnotes/heatwave/en/news-9-7-0.html#heatwave-9-7-0-hw-genai>
- MySQL HeatWave Documentation – NL_SQL
 - <https://dev.mysql.com/doc/heatwave/en/mys-hwgenai-nl-sql.html>
- Querying the Unstructured: Natural Language to SQL for JSON Data
 - <https://dasini.net/blog/2025/10/14/querying-the-unstructured-natural-language-to-sql-for-json-data/>
- Let Your AI DBA Assistant Write Your MySQL Queries
 - <https://dasini.net/blog/2025/10/07/let-your-ai-dba-assistant-write-your-mysql-queries/>
- Ask Your Database Anything: Natural Language to SQL in MySQL HeatWave
 - <https://dasini.net/blog/2025/09/30/ask-your-database-anything-natural-language-to-sql-in-mysql-heatwave/>

**Thank You to Our Community
Contributors!**

Thank You to Our Community Contributors!

- MySQL 9.7 includes contributions from community members helping improve:
 - Connector/NET
 - the MySQL Server Optimizer
 - Parser
 - Prepared Statements
 - Performance Schema
 - Types
 - Charsets
- The MySQL ecosystem continues to benefit from contributions across users, developers, DBAs, and partners worldwide

Thank You to Our Community Contributors!

We are grateful to the following contributors whose work was included in this release

<https://blogs.oracle.com/mysql/mysql-9-7-thank-you-for-your-contributions>

Thank You to Our Community Contributors!

- **Ma Liming** (Connector/NET)
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- **Yakir Gibraltar** (Optimizer)

Thanks for using MySQL!